UNIVERSITÀ DEGLI STUDI DI PISA

Corso di Studio in Bionics Engineering

Statistical Signal Processing - *Prova scritta del 05/11/2025*

PROBLEM

Let us consider the following model for the observed signal

$$z = x + \boldsymbol{y}^T \mathbf{1}_2, \tag{1}$$

where $x \sim \mathcal{N}(0,1)$ is a Gaussian signal, $\mathbf{1}_2 = [1,1]^T$, and

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^{2\times 1} \sim \mathcal{N}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}\right) \tag{2}$$

is the disturbance component such that (x, y_1, and y_2 are jointly Gaussian)

$$E[x(y_1 - 1)] = 0.5,\ E[x(y_2 - 2)] = 0.3 \tag{3}$$

Compute the following quantities:

1. the joint probability density function (PDF) of the vector $\boldsymbol{w} = [x, y_1, y_2]^T \in \mathbb{R}^{3\times 1}$;
2. the MMSE of x, its distribution, and the MSE (is it efficient?);
3. the maximum likelihood estimate of x (and its properties) assuming that x is deterministic and that K independent and identically distributed observations $z_1, \ldots, z_K$ are available.

SOLUTION

1. Since x, y_1, and y_2 are jointly Gaussian, vector $\boldsymbol{w}$ is Gaussian distributed with mean vector and covariance matrix

$$\boldsymbol{\mu}_w = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \text{ and } \boldsymbol{C}_w = \begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.5 & 1 & 0.5 \\ 0.3 & 0.5 & 1 \end{bmatrix}, \tag{4}$$

respectively. It follows that its PDF can be written as

$$\frac{1}{(2\pi)^{3/2}(0.56)^{1/2}} \exp\left\{ -\frac{1}{2} \left(\boldsymbol{w} - \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right)^T \begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.5 & 1 & 0.5 \\ 0.3 & 0.5 & 1 \end{bmatrix}^{-1} \left(\boldsymbol{w} - \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right) \right\}. \tag{5}$$

2. The MMSE of x is given by the conditional mean of x given $z = x + \boldsymbol{y}^T \mathbf{1}_2 = \boldsymbol{w}^T \mathbf{1}_3$, where $\mathbf{1}_3 = [1, 1, 1]^T$. To compute it, notice that z is a linear transformation of the Gaussian vector $\boldsymbol{w}$ and, hence, it is still Gaussian distributed with mean and variance given by

$$\eta_z = E[\boldsymbol{w}^T \mathbf{1}] = E[\boldsymbol{w}^T]\mathbf{1}_3 = [0, 1, 2]\mathbf{1}_3 = 3 \tag{6}$$

and

$$\begin{aligned} \sigma_z^2 &= E[(\boldsymbol{w}^T \mathbf{1}_3 - 3)^2] = E[(\boldsymbol{w}^T \mathbf{1}_3)^2] - 9 = \mathbf{1}_3^T E[\boldsymbol{w}\boldsymbol{w}^T]\mathbf{1}_3 - 9 \\ &= \mathbf{1}_3^T \left(\begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.5 & 1 & 0.5 \\ 0.3 & 0.5 & 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} [0, 1, 2] \right) \mathbf{1}_3 - 9 \\ &= \mathbf{1}^T \begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.5 & 2 & 2.5 \\ 0.3 & 2.5 & 5 \end{bmatrix} \mathbf{1} - 9 = 5.6, \end{aligned} \tag{7}$$

respectively. Moreover, the cross-covariance between x and z is

$$\begin{aligned} c_{xz} &= E[x(z - \mu_z)] = E[x(x + y_1 + y_2 - 1 - 2)] = E[x(x + (y_1 - 1) + (y_2 - 2))] \\ &= 1 + E[x(y_1 - 1)] + E[x(y_2 - 2)] = 1.8. \end{aligned} \tag{8}$$

Gathering the above results, we obtain

$$\hat{x}_{MMSE} = \frac{c_{xz}}{\sigma_z^2}(z - \eta_z) = \frac{1.8}{5.6}(z - 3), \tag{9}$$

that is Gaussian distributed with zero mean and variance $(1.8)^2/5.6$. Finally, the MSE is the conditional variance of x given z, namely

$$MSE(\hat{x}) = 1 - \frac{(1.8)^2}{5.6} = BCRB(\hat{x}). \tag{10}$$

3. Using the same line of reasoning as in the previous item, it is not difficult to show that $t = \boldsymbol{y}^T \mathbf{1}_2 \sim \mathcal{N}(3, 3)$ and, hence, $z_k \sim \mathcal{N}(3 + x, 3)$, $k = 1, \ldots, K$. The joint log-likelihood is

$$\mathcal{L}(x) = C - \frac{1}{6} \sum_{k=1}^{K} (z_k - 3 - x)^2, \tag{11}$$

where $C \in \mathbb{R}$ is a constant accounting for the terms independent of x. Notice that the objective function is a second order polynomial of x, thus we can find the stationary points by setting to zero its first derivative with respect to x, namely

$$\frac{\partial}{\partial x}\mathcal{L}(x) = \frac{1}{3}\sum_{k=1}^{K}(z_k - 3 - x) = 0 \tag{12}$$

$$\Rightarrow \sum_{k=1}^{K} z_k - 3K - Kx = 0 \tag{13}$$

$$\Rightarrow \hat{x}_{ML} = \frac{1}{K}\sum_{k=1}^{K} z_k - 3, \tag{14}$$

that is Gaussian with mean x (unbiased) and variance (that coincides with the MSE) $3/K$ (consistent and efficient).